

Project Verification Report

2021

COVER PAGE	
Project Verification Report Form (VR)	
BASIC INFORMATION	
Name of approved UCR Project Verifier / Reference No.	Enviance Services Private Limited
Type of Accreditation	☐ CDM or other GHG Accreditation ☒ ISO 14065 Accreditation
Approved UCR Scopes and GHG Sectoral scopes for Project Verification	01 Energy industries (Renewable/Non-Renewable Sources)
Validity of UCR approval of Verifier	30/09/2027
Completion date of this VR	15/07/2025
Title of the project activity	Vaayu India Wind Power Project in Jaisalmer, Rajasthan
Project reference no. (as provided by UCR Program)	UCR 501
Name of Entity requesting verification service (can be Project Owners themselves or any Entity having authorization of Project Owners, example aggregator.)	Viviid Emissions Reductions Universal Pvt. Ltd.
Contact details of the representative of the Entity, requesting verification service (Focal Point assigned for all communications)	Name: Lokesh Jain Email ID – lokesh.jain@viviidgreen.com
Country where project is located	India
Applied methodologies (approved methodologies by UCR Standard used)	ACM0002-Consolidated baseline methodology for grid-connected electricity generation from renewable sources - Version 22.0
GHG Sectoral scopes linked to the applied methodologies	01 Energy industries (Renewable/Non-Renewable Sources)

Project Verification Criteria: Mandatory requirements to be assessed	☒ UCR Standard ☒ Applicable Approved Methodology ☒ Applicable Legal requirements /rules of host country ☒ Eligibility of the Project Type ☒ Start date of the Project activity ☒ Meet applicability conditions in the applied methodology ☒ Credible Baseline ☒ Do No Harm Test ☒ Emission Reduction calculations ☒ Monitoring Report ☒ No GHG Double Counting ☐ Others (please mention below)
Project Verification Criteria: Optional requirements to be assessed	☒ Environmental Safeguards Standard and do-no-harm criteria ☒ Social Safeguards Standard do-no-harm criteria
Project Verifier's Confirmation: The <i>UCR Project Verifier</i> has verified the UCR project activity and therefore confirms the following:	The UCR Project Verifier Enviance Services Private Limited, certifies the following with respect to the UCR Project Activity Vaayu India Wind Power Project in Jaisalmer, Rajasthan ☒ The Project Owner has correctly described the Project Activity in the Project Concept Note 1.1 (dated 05/07/2025) including the applicability of the approved methodology <i>ACM0002-Consolidated baseline methodology for</i>

	grid-connected electricity generation from renewable sources - Version 22.0 and meets the methodology applicability conditions and has achieved the estimated GHG emission reductions, complies with the monitoring methodology and has calculated emission reductions estimates correctly and conservatively. ☒ The Project Activity is likely to generate GHG emission reductions amounting to the estimated 75,576 tCO_{2e} annually, as indicated in the PCN, which are additional to the reductions that are likely to occur in absence of the Project Activity and complies with all applicable UCR rules, including ISO 14064-2 and ISO 14064-3. ☒ The Project Activity is not likely to cause any net-harm to the environment and/or society ☒ The Project Activity complies with all the applicable UCR rules¹ and therefore recommends UCR Program to register the Project activity with above mentioned labels.
Project Verification Report, reference number and date of approval	Verification Report UCR Reference number: 501

¹https://a23e347601d72166dcd6-16da518ed3035d35cf0439f1cdf449c9.ssl.cf2.rackcdn.com//Documents/UCRtermsandconditionsMay2025Ver11_230525172325112351.pdf

	Date of approval: 15/07/2025
Name of the authorised personnel of UCR Project Verifier and his/her signature with date	 Vidhya Muralikrishna Quality Manager Date: 15/07/2025

PROJECT VERIFICATION REPORT

Executive summary

The project activity is titled- “Vaayu India Wind Power Project in Jaisalmer, Rajasthan”.

It is a wind-power Project which is spread across Sipla, Pithla, Kotri, Senag, Barna and Khuri villages of Jaisalmer District of Rajasthan State in India. The project consists of 63 machines of Enercon make E-53 type Wind Energy Converters (WECs) each of capacity 800 KW. The project has been effectively commissioned by Vaayu India Power Corporation Pvt Ltd. The project involves the supply, erection, commissioning, and operation of these machines, managed by Wind World (India) Limited (WWIL), which serves as the equipment supplier and Operation and Maintenance contractor.

The wind farm generates approximately **255611.099 MWh** of clean electricity in the current monitoring period. The generated electricity is supplied to Electricity Distribution Company (DISCOM), Jaipur Vidyut Vitran Nigam Limited, under a long-term power purchase agreement (PPA) signed between Vaayu (India) Power Corporation Private Limited (VIPCPL) and Jaipur Vidyut Vitran Nigam Limited. The expected operational lifetime of the project is for 20 years. Addressing the energy demand-supply gap in Rajasthan and supporting the region's sustainable growth.

The first WEC under the project activity was commissioned on 14/05/2011 and the last WEC under the project activity was commissioned on 14/07/2011. The project has been operational since the earliest commissioning date.

The project activity is registered under Clean Development Mechanism (CDM) project with registration number 5186ⁱ, as well as Gold Standard (GS) with reference number 5013ⁱⁱ. The crediting period of this project under CDM & GS was 01/10/2011-30/09/2021. PP seeks verification under UCR from 01/10/2021 onwards, i.e., crediting period for UCR starts from 01/10/2021. Hence, there is no double counting for the said project.

The project consists of 63 machines of Enercon make E-53 type Wind Energy Converters (WECs) each of capacity 800 KW.

Commissioning dates of the project activity are mentioned in the table below:

S. No.	WEG S.C. NO	No. & Capacity	Commissioning Date (dd/mm/yyyy)
1	2	1 X 800 kW	30-06-2011
2	3	1 X 800 kW	14-05-2011
3	4	1 X 800 kW	14-05-2011
4	5	1 X 800 kW	14-05-2011
5	6	1 X 800 kW	14-05-2011
6	7	1 X 800 kW	14-05-2011
7	8	1 X 800 kW	14-05-2011
8	9	1 X 800 kW	14-05-2011
9	10	1 X 800 kW	30-06-2011

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10	13	1 X 800 kW	14-05-2011
11	14	1 X 800 kW	14-05-2011
12	15	1 X 800 kW	14-05-2011
13	16	1 X 800 kW	14-05-2011
14	17	1 X 800 kW	14-05-2011
15	23	1 X 800 kW	14-05-2011
16	24	1 X 800 kW	14-05-2011
17	25	1 X 800 kW	14-05-2011
18	29	1 X 800 kW	14-05-2011
19	30	1 X 800 kW	14-05-2011
20	32	1 X 800 kW	14-05-2011
21	33	1 X 800 kW	14-05-2011
22	38	1 X 800 kW	14-07-2011
23	39	1 X 800 kW	14-07-2011
24	40	1 X 800 kW	14-07-2011
25	72	1 X 800 kW	14-07-2011
26	79	1 X 800 kW	14-05-2011
27	80	1 X 800 kW	14-05-2011
28	87	1 X 800 kW	14-05-2011
29	88	1 X 800 kW	14-05-2011
30	93	1 X 800 kW	30-06-2011
31	99	1 X 800 kW	15-05-2011
32	101	1 X 800 kW	14-07-2011
33	102	1 X 800 kW	14-07-2011
34	103	1 X 800 kW	15-05-2011
35	106	1 X 800 kW	15-05-2011
36	107	1 X 800 kW	15-05-2011
37	110	1 X 800 kW	15-05-2011
38	114	1 X 800 kW	15-05-2011
39	115	1 X 800 kW	15-05-2011
40	117	1 X 800 kW	15-05-2011
41	118	1 X 800 kW	15-05-2011
42	123	1 X 800 kW	15-05-2011
43	124	1 X 800 kW	15-05-2011
44	125	1 X 800 kW	15-05-2011
45	144	1 X 800 kW	14-05-2011
46	148	1 X 800 kW	14-05-2011
47	150	1 X 800 kW	14-05-2011
48	151	1 X 800 kW	30-06-2011
49	152	1 X 800 kW	14-05-2011

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50	157	1 X 800 kW	14-07-2011
51	158	1 X 800 kW	30-06-2011
52	159	1 X 800 kW	30-06-2011
53	503	1 X 800 kW	15-05-2011
54	504	1 X 800 kW	15-05-2011
55	505	1 X 800 kW	15-05-2011
56	507	1 X 800 kW	14-05-2011
57	508	1 X 800 kW	14-05-2011
58	512	1 X 800 kW	30-06-2011
59	513	1 X 800 kW	30-06-2011
60	514	1 X 800 kW	14-07-2011
61	520	1 X 800 kW	15-05-2011
62	521	1 X 800 kW	15-05-2011
63	525	1 X 800 kW	15-05-2011

Geo Co-ordinates of the project activity are mentioned in the table below:

Sr.No	Loc.No.	Village	Latitude			Longitude		
			Deg.	Minute	Second	Deg.	Minute	Second
1	2	Kuhri	26	40	25.4	70	44	21.9
2	3	Barna	26	40	33.2	70	44	18.6
3	4	Barna	26	40	33.8	70	44	11.2
4	5	Barna	26	40	27.2	70	43	58
5	6	Barna	26	40	39.7	70	43	57.1
6	7	Barna	26	40	35.8	70	43	47.2
7	8	Barna	26	40	35.5	70	43	39.6
8	9	Barna	26	40	58	70	44	3.5
9	10	Sipla	26	41	14.8	70	44	27.3
10	110	Senag	26	42	0.7	70	48	28.8
11	13	Barna	26	41	48.8	70	44	20.1
12	14	Barna	26	41	49.3	70	44	11.2
13	15	Barna	26	40	45.6	70	43	31.6
14	16	Barna	26	40	43.6	70	43	24.2
15	17	Barna	26	41	0.4	70	43	26.1
16	23	Barna	26	41	37	70	42	50.3
17	24	Barna	26	41	44.7	70	42	44.7
18	25	Barna	26	41	49.2	70	42	37.1
19	29	Barna	26	42	5.7	70	42	27.3
20	30	Barna	26	42	9.7	70	42	37.7
21	32	Barna	26	42	18.6	70	42	52.6

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22	33	Barna	26	42	18.7	70	43	10.8
23	38	Sipla	26	42	24.6	70	43	28.8
24	39	Sipla	26	42	23.7	70	43	36.8
25	40	Sipla	26	42	22.9	70	43	44.9
26	72	Khuhri	26	41	53.3	70	45	50.8
27	79	Pithla	26	42	40.3	70	46	11.5
28	80	Pithla	26	42	44.6	70	46	3.2
29	87	Pithla	26	42	53.1	70	46	24.9
30	88	Kotri	26	42	43.1	70	46	31.3
31	93	Senag	26	42	18.4	70	46	55.6
32	99	Senag	26	42	0.9	70	47	39
33	101	Senag	26	42	27.4	70	47	39
34	102	Senag	26	42	33.1	70	47	30.8
35	103	Kotri	26	42	22.5	70	47	49.1
36	106	Kotri	26	42	25.4	70	48	15.2
37	107	Kotri	26	42	16.8	70	48	22
38	114	Senag	26	41	39	70	48	6.1
39	115	Kotri	26	41	42.2	70	48	16.2
40	117	Kotri	26	41	35.3	70	48	31
41	118	Senag	26	41	36.4	70	48	39.1
42	123	Pithla	26	42	19.5	70	49	11.2
43	124	Pithla	26	42	2	70	49	10.1
44	125	Pithla	26	41	54.2	70	49	15.6
45	144	Pithla	26	43	43.5	70	45	56.4
46	148	Pithla	26	44	3.9	70	46	3.1
47	150	Pithla	26	44	21	70	46	8.5
48	151	Pithla	26	44	20.9	70	45	59
49	152	Pithla	26	44	32.9	70	46	22.6
50	157	Pithla	26	44	48.7	70	46	4.5
51	158	Pithla	26	44	36	70	45	52
52	159	Pithla	26	44	40.9	70	45	46.8
53	503(S)	Pithla	26	41	11.5	70	48	38.9
54	504(S)	Pithla	26	41	19.3	70	48	35
55	505(S)	Kotri	26	41	19.3	70	48	27.4
56	507(S)	Kotri	26	42	20.7	70	46	34.8
57	508(S)	Kotri	26	42	22.8	70	46	27.1
58	512(S)	Kotri	26	42	40.8	70	47	1.9
59	513(S)	Kotri	26	42	35.2	70	47	9
60	514(S)	Kotri	26	42	32	70	47	21.9
61	520(S)	Senag	26	42	44.3	70	48	7.3
62	521(S)	Senag	26	42	38.2	70	48	19.2
63	525(S)	Senag	26	41	50.6	70	49	23.5

Proposed wind power project has evolved as a result of the policies of Government of India and Government of Rajasthan, which encourages energy development from renewable sources. These policies have given fresh impetus to wind power generation.

The Project Activity is a greenfield wind project and the generated electricity is supplied to Electricity Distribution Company (DISCOM), Jaipur Vidyut Vitran Nigam Limited. Addressing the energy demand-supply gap in Rajasthan and supporting the region's sustainable growth. A Power Purchase Agreement has been signed between Vaayu (India) Power Corporation Private Limited (VIPCL) and Jaipur Vidyut Vitran Nigam Limited. The project activity involves 63 numbers wind energy converters (WECs) of Enercon make (800 KW, E53) with internal electrical lines connecting the project activity with local evacuation facility. The 50.4 MW wind power project involves 63 E-53 Wind Energy Converters (WECs), each with an 800-kW capacity, set up by WWIL in India. These wind turbines convert wind energy into electricity using synchronous generators, which are manufactured at WWIL's is spread across villages of Rajasthan in Jaisalmer district. These machines generate electricity that is stepped up from 400 V to 33 kV and fed into the Rajasthan state electricity grid, which is part of the interconnected Indian grid. Plant using advanced vacuum impregnation technology for better insulation and durability. The turbines have rotor blades, a nacelle with the generator and control systems, a tower, and a concrete foundation. The electricity generated is fed into the Indian grid through transformers. Without this project, the same amount of electricity would come from fossil fuel-based power plants, which is the baseline scenario. This renewable energy project reduces emissions and supports local manufacturing through technology transfer.

The WECs generate 3-phase power at 400V, which is stepped up to 33 KV. The project activity can operate in the frequency range of 48.5–51.5 Hz and in the voltage range of $400\text{ V} \pm 12.5\%$. The average lifetime of the WEG is around 20 years as per the industry standards. As per DPR plant load factor is of 21.40%. The project being a renewable energy generation activity, leads to reduction in fossil fuel dominated electricity generation from the Indian grid.

The core objective of this project activity is to displace an equivalent amount of electricity which would have otherwise been generated by fossil fuel dominant electricity grid. The estimated lifetime of the project activity is considered as 20 years for wind technology. In the Pre- project scenario the entire electricity, consumed by the customers or delivered to the grid by, would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.

The project activity consists of 63 Wind turbines of 800kW manufactured and supplied by Enercon. This project generates 50.4 MW power which is supplied to the state electricity utility delivered by the Project Proponent. The applied technology is one of the most environment friendly technologies available as the operation of the wind power plant does not emit any GHGs or any other harmful gases unlike the operation of conventional power plant. The project activity has used the reliable and proven technology to ensure that an environmentally safe and sound technology has been implemented.

The main component of this project activity is wind turbine which consists of components like main tower, blades, nacelle, hub, main shaft, gear box, bearing and housing, brake and generator. The generation of power from wind turbines is a clean technology as there is no fossil fuel-fired or no GHG gases are emitted during the process. Thus, project activity leads to a reduction the GHG emissions as it displaces power from fossil fuel-based electricity generation in the regional grid. Since the project activity generates electricity through wind energy, it will not cause any negative impact on the environment and thereby contributes to climate change mitigation efforts.

The project activity also contributes to SDG goals 7,8 and 13.

The first crediting period of the project activity in UCR is 03 years, 03 months in which total estimated electricity generation is 94,481.86 MWh annually and the total GHG emission reduction estimated is 75,576 tCO₂e annually.

The electricity generation for the current monitoring period is 2,55,611 MWh and total GHG emission reduction is 2,20,152 tCO₂e.

Scope of Verification

The scope of the services for the project is to perform Project Verification of concerned Project Activity. The scope of verification is to assess the claims and assumptions made in the Project Concept Note (PCN) and Monitoring Report (MR) against the UCR criteria, including but not limited to, UCR program verification guidance document, UCR Standard, UCR Program Manual, and related rules and guidelines established under Program process.

Verification Process and Methodology

The verification process was undertaken by a competent verification team and involved the following,

- Desk review of documents and evidence submitted in context of the reference rules and guidelines issued by UCR,
- Undertaking/conducting site visit/remote audit, interview or interactions with the representative of the project owners/representatives,
- Reporting audit findings with respect to clarifications and non-conformities and the closure of the findings, as appropriate and preparing a draft verification opinion based on the auditing findings and conclusions
- Finalization of the verification opinion (this report)

Desk/Document review

A detailed desk review of the PCN, MR, Methodology and all other associated documentation and references took place in advance of the site visit, and additional documents that were not available for the desk review were requested for review during the site visit. Additional information can be required to complete the verification, which may be obtained from other public and reliable sources or through telephone and face to face interviews with key

stakeholders (including the project developers and where necessary, government and NGO representatives in the host country).

A list of all documents reviewed or referred to in the course of this verification is included in Appendix 3 below.

Follow up interviews/site visit

The verifier conducted remote audit and had requested for site photographs, short videos. A remote interview was conducted with the project owners and stakeholders.

Conclusion

Based on the work performed, the verifier concludes that in the project activity “Vaayu India Wind Power Project in Jaisalmer, Rajasthan”, the information and data presented in the MR version 1.1 dated 05/07/2025 is in line with the Project Concept Note Version 1.1 date 05/07/2025 and meets all relevant requirements of the UCR for UCR project activities. The UCR project activity correctly applies the methodology “ACM0002-Consolidated baseline methodology for grid-connected electricity generation from renewable sources -Version 22.0” leading to result in real, measurable and long-term emission reductions achieved for the current monitoring period.

For the current monitoring period, verified emission reductions achieved by the project activity were as below;

Start date of monitoring period	01/10/2021
End date of monitoring period	31/12/2024
Emission reductions achieved	2,20,152 tCO ₂ eq

Project Verification team, technical reviewer and approver

Project Verification team

No.	Role	Last name	First name	Affiliation (e.g. name of central or other office of UCR Project Verifier or outsourced entity)	Involvement in		
					Doc review	Off-Site inspection	Interviews
1.	Team Leader/ Technical Expert	Singh	Ritu	Enviance Services Private Limited	Yes	Yes	Yes
2.	Team Leader in Trainee	Mahajan	Swati	Enviance Services Private Limited	Yes	Yes	Yes

Technical reviewer and approver of the Project Verification report

No.	Role	Type of resource	Last name	First name	Affiliation (e.g. name of central or other office of UCR Project Verifier or outsourced entity)
1.	Technical reviewer	Internal	Kumar	Mr. Pankaj	Enviance Services Private Limited
2.	Approver	Internal	Krishna	Vidhya Murali	Enviance Services Private Limited

Means of Project Verification

Desk/document review

A detailed desk review of the PCN, MR, methodology and all other associated documentation and references took place in advance of the remote audit, and additional documents that were not available for the desk review were requested for review during the remote audit. Additional information can be required to complete the verification, which may be obtained from other public and reliable sources or through telephone and face-to face interviews with key stakeholders (including the project developers and where necessary, Government and NGO representatives in the host country).

A list of all documents reviewed or referred to in the course of this verification is included in Appendix 3 below.

Off-site inspection

Date of Off-site Inspection: 01/07/2025			
No.	Activity performed Off-Site	Site location	Date
1.	a) An assessment of the implementation and operation of the project activity as per the PCN and UCR requirements b) Verification of the project design, as documented is sound and reasonable, and meets the identified criteria of UCR Standard Requirements and associated guidance c) Assessment to conformance with the certification criteria as laid out in the UCR Standards; d) Evaluation of the conformance with the certification scope, including the GHG project and baseline scenarios, additionality; GHG sources, sinks, and reservoirs; and the physical infrastructure, activities, technologies and processes of the GHG	Sipla, Pithla, Kotri, Senag, Barna and Khuri villages of Jaisalmer District of Rajasthan State in India.	01/07/2025

	project to the requirements of the UCR; e) Evaluation of the calculation of GHG emissions, including the correctness and transparency of formulae and factors used; assumptions related to estimating GHG emission reductions; and uncertainties; and determination whether the project could reasonably be expected to achieve the estimated GHG reduction/removals. f) Review of information flows for generating, aggregating and reporting of the parameters to be monitored g) To confirm that the operational and data collection procedures can be implemented in accordance with the Monitoring Plan h) Cross-check of information provided in the submitted documents and data from other sources available at site i) Review of calculations and assumptions made in determining the GHG data and estimated ERs, and an identification of QA/QC procedures in place to prevent, or identify and correct, any errors or omissions in the reported monitoring parameters Interviews of local Stakeholders		
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Interviews

No.	Interview			Date	Subject
	Last name	First name	Affiliation		
1.	Kummawat	Lal Chand	Vaayu India Power Corporation Pvt Ltd.	01/07/2025	Project Implementation, Monitoring plan, Project Boundary, Eligibility criteria, Host country requirements, Emission reduction calculations Project implementation, monitoring, Local stakeholder consultation
2.	Girigosavi	Tushar	Viviid emissions		
3.	Darne	Minal	reductions universal private Ltd.		
4.	Raj	Tilak	Local Stakeholders		
5.	Kumar	Manas			
6.	Kumar	Rakesh			
7.	Kurmi	Rahul			

Sampling approach

Not Applicable.

Clarification request (CLs), corrective action request (CARs) and forward action request (FARs) raised

Areas of Project Verification findings	No. of CL	No. of CAR	No. of FAR
Green House Gas (GHG)			
Identification and Eligibility of project type	-	-	-
General description of project activity	02	-	-
Application and selection of methodologies and standardized baselines	-	-	-
- Application of methodologies and standardized baselines	-	-	-
- Deviation from methodology and/or methodological tool	-	-	-
- Clarification on applicability of methodology, tool and/or standardized baseline	01	01	-
- Project boundary, sources and GHGs	-	-	-
- Baseline scenario	-	-	-
- Estimation of emission reductions or net anthropogenic removals	-	-	-
- Monitoring Report	-	01	-
Start date, crediting period and duration	-	01	-
Environmental impacts	-	-	-
Project Owner- Identification and communication	-	-	-
Others (please specify)	01	-	-
Total	04	03	-

Project Verification findings

Identification and eligibility of project type

Means of Project Verification	The project has an installation of a 50.4 MW (0.8 MW x 63) wind power capacity and hence it qualifies as a large-scale project. This is confirmed based on the commissioning certificates and technical specifications. Since the project is a large-scale project, it has applied approved CDM large scale methodology ACM0002-Consolidated baseline methodology for grid-connected electricity generation from renewable sources -Version 22.0. The Project owner has used valid MR form available at the UCR website for the preparation of MR for the current project activity. The project has prepared MR in line with UCR guidance and requirements.
Findings	No findings raised.
Conclusion	The UCR-approved format is used for description and the project meets the requirement of the UCR verification standard and UCR project standard. UCR project communication agreement was submitted to the verifier and the same has been verified. Methodology referenced and applied appropriately describing the project type. The eligibility of the project aggregator is verified using the UCR communication agreement, project correctly applies the verification standard, UCR project standard, and UCR regulations. The project activity is overall meeting the requirements of the UCR Verification standard and UCR project standard.

General description of project activity

Means of Project Verification

The project activity involves the operation of a 50.4 MW (0.8 MW x 63) of large-scale wind power project and its commissioning date and power evacuation at the substation were verified through the commissioning certificate of the project. The power purchase agreement confirms the companies/entities involved in the agreement for purchase of electricity from the 50.4 MW (Sipla, Pithla, Kotri, Senag, Barna and Khuri villages of Jaisalmer District of Rajasthan State in India.) project.

Assessment team conducted documentation review of the PCN against the UCR program verification standard version 2.0 and UCR CoU Standard (project eligibility criteria) version 7.0 and the UCR-PCN-FORM Version 1.0.

By checking the supporting documents, it is confirmed that the project is a greenfield wind power project, the project is located in Sipla, Pithla, Kotri, Senag, Barna and Khuri villages of Jaisalmer District of Rajasthan State in India. The approximate geo-coordinates of the project locations are mentioned below.

Details of Latitude & Longitude for the project site: -

Sr.No	Loc.No.	Village	Latitude			Longitude		
			Deg.	Minute	Second	Deg.	Minute	Second
1	2	Khuhri	26	40	25.4	70	44	21.9
2	3	Barna	26	40	33.2	70	44	18.6
3	4	Barna	26	40	33.8	70	44	11.2
4	5	Barna	26	40	27.2	70	43	58
5	6	Barna	26	40	39.7	70	43	57.1
6	7	Barna	26	40	35.8	70	43	47.2
7	8	Barna	26	40	35.5	70	43	39.6
8	9	Barna	26	40	58	70	44	3.5
9	10	Sipla	26	41	14.8	70	44	27.3
10	110	Senag	26	42	0.7	70	48	28.8
11	13	Barna	26	41	48.8	70	44	20.1
12	14	Barna	26	41	49.3	70	44	11.2
13	15	Barna	26	40	45.6	70	43	31.6
14	16	Barna	26	40	43.6	70	43	24.2
15	17	Barna	26	41	0.4	70	43	26.1
16	23	Barna	26	41	37	70	42	50.3
17	24	Barna	26	41	44.7	70	42	44.7
18	25	Barna	26	41	49.2	70	42	37.1
19	29	Barna	26	42	5.7	70	42	27.3
20	30	Barna	26	42	9.7	70	42	37.7
21	32	Barna	26	42	18.6	70	42	52.6

	22	33	Barna	26	42	18.7	70	43	10.8
	23	38	Sipla	26	42	24.6	70	43	28.8
	24	39	Sipla	26	42	23.7	70	43	36.8
	25	40	Sipla	26	42	22.9	70	43	44.9
	26	72	Khuhri	26	41	53.3	70	45	50.8
	27	79	Pithla	26	42	40.3	70	46	11.5
	28	80	Pithla	26	42	44.6	70	46	3.2
	29	87	Pithla	26	42	53.1	70	46	24.9
	30	88	Kotri	26	42	43.1	70	46	31.3
	31	93	Senag	26	42	18.4	70	46	55.6
	32	99	Senag	26	42	0.9	70	47	39
	33	101	Senag	26	42	27.4	70	47	39
	34	102	Senag	26	42	33.1	70	47	30.8
	35	103	Kotri	26	42	22.5	70	47	49.1
	36	106	Kotri	26	42	25.4	70	48	15.2
	37	107	Kotri	26	42	16.8	70	48	22
	38	114	Senag	26	41	39	70	48	6.1
	39	115	Kotri	26	41	42.2	70	48	16.2
	40	117	Kotri	26	41	35.3	70	48	31
	41	118	Senag	26	41	36.4	70	48	39.1
	42	123	Pithla	26	42	19.5	70	49	11.2
	43	124	Pithla	26	42	2	70	49	10.1
	44	125	Pithla	26	41	54.2	70	49	15.6
	45	144	Pithla	26	43	43.5	70	45	56.4
	46	148	Pithla	26	44	3.9	70	46	3.1
	47	150	Pithla	26	44	21	70	46	8.5
	48	151	Pithla	26	44	20.9	70	45	59
	49	152	Pithla	26	44	32.9	70	46	22.6
	50	157	Pithla	26	44	48.7	70	46	4.5
	51	158	Pithla	26	44	36	70	45	52
	52	159	Pithla	26	44	40.9	70	45	46.8
	53	503(S)	Pithla	26	41	11.5	70	48	38.9
	54	504(S)	Pithla	26	41	19.3	70	48	35
	55	505(S)	Kotri	26	41	19.3	70	48	27.4
	56	507(S)	Kotri	26	42	20.7	70	46	34.8
	57	508(S)	Kotri	26	42	22.8	70	46	27.1
	58	512(S)	Kotri	26	42	40.8	70	47	1.9
	59	513(S)	Kotri	26	42	35.2	70	47	9
	60	514(S)	Kotri	26	42	32	70	47	21.9
	61	520(S)	Senag	26	42	44.3	70	48	7.3
	62	521(S)	Senag	26	42	38.2	70	48	19.2
	63	525(S)	Senag	26	41	50.6	70	49	23.5

	Assessment team performed an offsite inspection of project and confirmed that the location described in the PCN are accurate. The Project is a wind power project, to utilize wind energy to generate zero carbon emission electricity which is mainly dominated by fossil fuel power output. The project includes integrated power transmission mechanism, high performance rotor blades, dual speed asynchronous generator, microprocessor based fully automatic control system with user friendly operation and central monitoring system. Quality, Safety and Health plan for construction, installation, commissioning and Operation & Maintenance. Microprocessor controlled high efficiency soft start. Active Yaw gear drives incorporating hydraulic yaw brakes.
Findings	CL 01 and CL 04 were raised and closed successfully. More information presented in the appendix below.
Conclusion	The description of the project activity is verified to be true based on the review of PCN, MR, Commissioning Certificate and power purchase agreement.

Application and selection of methodologies and standardized baselines

(.a.i) Application of methodology and standardized baselines

Means of Project Verification	The project has taken the reference of CDM methodology ACM0002-Consolidated baseline methodology for grid-connected electricity generation from renewable sources -Version 22.0. CDM website is referred to check the latest version of the methodology. For the applicability mentioned in the PCN and MR, technical Specification, and commissioning certificate.
Findings	No findings raised.
Conclusion	The methodology applied is appropriately meeting the requirements of UCR and its standardized baseline. The methodology version is correct and valid. The referenced methodology is applicable to project activity.

(.a.ii) Clarification on applicability of methodology, tool and/or standardized baseline

Means of Project Verification	The documents reviewed are CDM methodology ACM0002-Consolidated baseline methodology for grid-connected electricity generation from renewable sources -Version 22.0, UCR Program standard, and UCR Verification Standard.
Findings	CL 03 and CAR 02 were raised and closed successfully. More information presented in the appendix below.
Conclusion	The verification team confirms that all the applicability criteria set by the applied CDM methodology and its eligible tools are met. The relevant information against those criteria is also included in the PCN and MR Ver.1.1. The selected CDM methodology for the project activity is applicable.

(.a.iii) Project boundary, sources and GHGs

Means of Project Verification	Project owner has considered project boundary as per applicable
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	methodology ACM0002-Consolidated baseline methodology for grid-connected electricity generation from renewable sources -Version 22.0, "the spatial extent of this project activity includes the project site and all the power plants connected physically to the electricity system (grid) that the power project is connected to." Review of PCN and MR confirms that project sites and Indian electricity grid system is considered as a project boundary which is appropriate.
Findings	No findings raised
Conclusion	The project boundary is correctly defined in the PCN and MR. GHGsources are correctly identified and reported. The project meets the requirements of UCR project standard, Verification standard and methodology requirements for a boundary, GHG sources.

(.a.iv) **Baseline scenario**

Means of Project Verification	As per the applied ACM0002-Consolidated baseline methodology for grid-connected electricity generation from renewable sources - Version 22.0 the baseline scenario is as following: The baseline scenario is that if the project activity is the installation of a Greenfield power plant, the baseline scenario is electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources to the grid. Remote audit conducted and document review showed that in absence of the project activity, the generated electricity would have been supplied by the Indian grid which is dominated by fossil fuel fired plants.
Findings	No findings raised.
Conclusion	The approved baseline methodology has been correctly applied to identify a realistic and credible baseline scenario, and the identified baseline scenario most reasonably represents what would occur in the absence of the proposed UCR project activity. All the assumption and data used by the project participants are listed in the PCN and/or supporting documents. All documentation relevant for establishing the baseline scenario are correctly quoted and interpreted in the PCN. Assumptions and data used in the identification of the baseline scenario are justified appropriately, supported by evidence and can be deemed reasonable.

(.a.v) **Estimation of emission reductions or net anthropogenic removal**

Means of Project Verification	The project verification team checked whether the equations and parameters used to calculate GHG emission reductions or net anthropogenic GHG removals for PCN and MR are in accordance with applied methodology. Project verification team checked section B.5 and C.5.1 of the PCN & MR respectively to confirm whether all formulae to calculate baseline emissions, project emission and leakage have been applied in line with the underlying methodology. The emission reduction calculation has been carried out as per the CDM methodology ACM0002-Consolidated baseline methodology for grid-connected electricity generation from renewable sources -Version 22.0 As per the CDM approved ACM0002-Consolidated baseline methodology for grid-connected electricity generation from renewable sources -Version 22.0 paragraph 57, encompass solely the CO₂ emissions stemming from electricity generation in power plants displaced by the project activity. The methodology operates on the assumption that any electricity generation exceeding baseline levels would have originated from established grid-connected power plants and the integration of new grid-connected power plants. $BE_y = EG_{PJ, y} \times EF_{grid, CM, y}$ Where; BE_y: Baseline emissions in year y (tCO₂/year) EG_{PJ, y}: Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the UCR project activity in year y (MWh/year) EF_{grid, CM, y}: Combined margin CO₂ emission factor for grid connected power generation in year y (tCO₂/MWh) A "grid emission factor" refers to a CO₂ emission factor (tCO₂/MWh) which will be associated with each unit of electricity provided by an electricity system. The UCR recommends an emission factor of 0.9 tCO₂/MWh for the 2013-2023 years as a fairly conservative estimate for Indian projects not previously verified under any GHG program. Also, for the vintage 2021, the combined margin emission factor calculated from CEA database in India results into higher emission than the default value. Hence, the same emission factor has been considered to calculate the emission reduction under conservative approach.² Similarly, for the year 2024, a grid emission factor of 0.757 tCO₂/MWh is to be applied. These conservative factors are used to calculate emission reductions.
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² https://a23e347601d72166dcd6-16da518ed3035d35cf0439f1cdf449c9.ssl.cf2.rackcdn.com/Documents/UCRStandardAug2024updatedVer7_020824191534797526.pdf

	In order to facilitate adoption of authentic baseline emissions data and in keeping with the principle of “conservativeness,” all UCR Indian RE projects shall use the new conservative grid emission factor of 0.757 tCO₂/MWh in their emission reduction calculations for the 2024 vintage year. https://medium.com/@UniversalCarbonRegistry/ucr-cou-standard-update-2024-vintage-ucr-indian-grid-emission-factor-announced-ddb790cdc603 Project emissions: Regarding project emissions, ACM0002 version 22.0 specifies that only emissions related to fossil fuel combustion, emissions from the operation of geothermal power plants due to the release of non-condensable gases, and emissions from water reservoirs of hydroelectric plants should be taken into account. Since the project involves a wind power project, emissions from renewable energy plants are negligible $PE_y = 0$. Since wind power is a GHG emission free source of energy project emission considered as Zero for the project activity. Leakage Emissions: Leakage, as outlined in ACM0002 version 22.0, para 5.6, is considered to be zero as there is no transfer of energy-generating equipment in the project activity Hence ($LE_y = 0$). Emission reductions: As per approved ACM0002-Consolidated baseline methodology for grid-connected electricity generation from renewable sources -Version 22.0, emission reduction is estimated as difference between the baseline emission and project emission after factoring into leakage. Thus, $ER_y = BE_y - PE_y - LE_y$ Where: <table> <tr> <td>ER_y</td><td>= Emission reductions in year y (t CO₂)</td></tr> <tr> <td>BE_y</td><td>= Baseline Emissions in year y (t CO₂)</td></tr> <tr> <td>PE_y</td><td>= Project emissions in year y (t CO₂)</td></tr> <tr> <td>LE_y</td><td>= Leakage emissions in year y (t CO₂)</td></tr> </table> Therefore, $ER_y = BE_y$ The earliest commissioning date of the Project is 14/05/2011 when the first installation of the wind turbine was done and the last commissioning date is 14/07/2011. The start date of the crediting period under UCR is considered from 01/10/2021. The project activity is registered under Clean Development Mechanism (CDM) project with registration number 5186, as well as Gold Standard (GS) with reference number 5013. The crediting period of this project under CDM	ER_y	= Emission reductions in year y (t CO ₂)	BE_y	= Baseline Emissions in year y (t CO ₂)	PE_y	= Project emissions in year y (t CO ₂)	LE_y	= Leakage emissions in year y (t CO ₂)
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LE_y	= Leakage emissions in year y (t CO ₂)								

	& GS is 01/10/2011 to 30/09/2021.																																																																																																						
	For the ease of the calculation, duration of the crediting period in UCR is started from 01/10/2021 to 31/12/2024.																																																																																																						
	The estimated emission reductions are 75,576 CoUs/yr (75,576 tCO2eq/yr)																																																																																																						
	<table><tr><th rowspan="2">Year</th><th>Net Generation</th><th>Baseline Emissions</th><th>Project Emissions</th><th>Leakage</th><th>EF</th><th>Emission Reductions</th></tr><tr><th>MWh</th><th>(tCO2e)</th><th>(tCO2e)</th><th>(tCO2e)</th><th>(tCO2/MWh)</th><th>(tCO2e)</th></tr><tr><td>Year 1</td><td>94481.86</td><td>85033.67</td><td>0.00</td><td>0.00</td><td>0.9</td><td>85033.67</td></tr><tr><td>Year 2</td><td>94481.86</td><td>85033.67</td><td>0.00</td><td>0.00</td><td>0.9</td><td>85033.67</td></tr><tr><td>Year 3</td><td>94481.86</td><td>85033.67</td><td>0.00</td><td>0.00</td><td>0.9</td><td>85033.67</td></tr><tr><td>Year 4</td><td>94481.86</td><td>71522.76</td><td>0.00</td><td>0.00</td><td>0.757</td><td>71522.76</td></tr><tr><td>Year 5</td><td>94481.86</td><td>71522.76</td><td>0.00</td><td>0.00</td><td>0.757</td><td>71522.76</td></tr><tr><td>Year 6</td><td>94481.86</td><td>71522.76</td><td>0.00</td><td>0.00</td><td>0.757</td><td>71522.76</td></tr><tr><td>Year 7</td><td>94481.86</td><td>71522.76</td><td>0.00</td><td>0.00</td><td>0.757</td><td>71522.76</td></tr><tr><td>Year 8</td><td>94481.86</td><td>71522.76</td><td>0.00</td><td>0.00</td><td>0.757</td><td>71522.76</td></tr><tr><td>Year 9</td><td>94481.86</td><td>71522.76</td><td>0.00</td><td>0.00</td><td>0.757</td><td>71522.76</td></tr><tr><td>Year 10</td><td>94481.86</td><td>71522.76</td><td>0.00</td><td>0.00</td><td>0.757</td><td>71522.76</td></tr><tr><td>Total Emission reduction</td><td>944818</td><td>755760</td><td>0</td><td>0</td><td></td><td>755760</td></tr><tr><td>Average Emission Reduction</td><td>94482</td><td>75576</td><td>0</td><td>0</td><td></td><td>75,576</td></tr></table>						Year	Net Generation	Baseline Emissions	Project Emissions	Leakage	EF	Emission Reductions	MWh	(tCO2e)	(tCO2e)	(tCO2e)	(tCO2/MWh)	(tCO2e)	Year 1	94481.86	85033.67	0.00	0.00	0.9	85033.67	Year 2	94481.86	85033.67	0.00	0.00	0.9	85033.67	Year 3	94481.86	85033.67	0.00	0.00	0.9	85033.67	Year 4	94481.86	71522.76	0.00	0.00	0.757	71522.76	Year 5	94481.86	71522.76	0.00	0.00	0.757	71522.76	Year 6	94481.86	71522.76	0.00	0.00	0.757	71522.76	Year 7	94481.86	71522.76	0.00	0.00	0.757	71522.76	Year 8	94481.86	71522.76	0.00	0.00	0.757	71522.76	Year 9	94481.86	71522.76	0.00	0.00	0.757	71522.76	Year 10	94481.86	71522.76	0.00	0.00	0.757	71522.76	Total Emission reduction	944818	755760	0	0		755760	Average Emission Reduction	94482	75576	0	0		75,576
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The actual emission reduction achieved during the first CoU's period (01/10/2021 to 31/12/2024) as per the Project Activity:																																																																																																							
Actual Total baseline emission reductions (BEy)= 2,20,152 CoUs (2,20,152 tCO2eq)																																																																																																							
<table><tr><th rowspan="2">Sr. No</th><th rowspan="2">Month & Year</th><th>Net Generation</th><th>Emission Factor</th><th>Baseline Emissions</th><th>Emission Reductions</th></tr><tr><th>MWh</th><th>(tCO2e/MWh)</th><th>(tCO2e)</th><th>tCO2e</th></tr><tr><td>1</td><td>01/10/2021 to 31/12/2021</td><td>13482.196</td><td>0.9</td><td>12132</td><td>12132</td></tr><tr><td>2</td><td>01/01/2022 to 31/12/2022</td><td>83241.344</td><td>0.9</td><td>74910</td><td>74910</td></tr><tr><td>3</td><td>01/01/2023 to 31/12/2023</td><td>89831.826</td><td>0.9</td><td>80842</td><td>80842</td></tr><tr><td>4</td><td>01/01/2024 to 31/12/2024</td><td>69055.733</td><td>0.757</td><td>52268</td><td>52268</td></tr><tr><td colspan="2">Total Emissions Reductions</td><td>255611.099</td><td colspan="2"></td><td>2,20,152</td></tr></table>						Sr. No	Month & Year	Net Generation	Emission Factor	Baseline Emissions	Emission Reductions	MWh	(tCO2e/MWh)	(tCO2e)	tCO2e	1	01/10/2021 to 31/12/2021	13482.196	0.9	12132	12132	2	01/01/2022 to 31/12/2022	83241.344	0.9	74910	74910	3	01/01/2023 to 31/12/2023	89831.826	0.9	80842	80842	4	01/01/2024 to 31/12/2024	69055.733	0.757	52268	52268	Total Emissions Reductions		255611.099			2,20,152																																																										
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Findings	No findings raised.																																																																																																						
Conclusion	In summary, the calculation of emission reductions was correctly demonstrated by the PP according to the methodology ACM0002-Consolidated baseline methodology for grid-connected electricity																																																																																																						

	generation from renewable sources -Version 22.0. It is confirmed by the assessment team that: (a) All assumptions made for estimating GHG are listed in the PCN; (b) All documentation used by the project participants as the basis for assumptions and source of data is correctly quoted and interpreted in the PCN (c) All values used in the PCN including GWPs are considered reasonable in the context of the proposed UCR project activity; (d) The methodologies and, where applicable, the standardized baselines and the other methodological regulatory documents have been applied correctly to calculate baseline, project and leakage GHG emissions, as well as GHG emission reductions; (e) All estimates of the baseline GHG emissions can be replicated using the data and parameter values provided in the PCN;
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(.a.vi) Monitoring Report

Means of Project Verification	Parameters determined- Ex-ante The following parameters are determined ex-ante and verified by the verification team: The baseline emission factor ($EF_{grid, y}$) of the project is reported to be determined ex-ante and would remain fixed for the crediting period. A "grid emission factor" refers to a CO₂ emission factor (tCO₂/MWh) which will be associated with each unit of electricity provided by an electricity system. The UCR recommends an emission factor of 0.9 tCO₂/MWh for the 2013-2023 years as a fairly conservative estimate for Indian projects not previously verified under any GHG program. Also, for the vintage 2021, the combined margin emission factor calculated from CEA database in India results into higher emission than the default value. Similarly, for the year 2024, a grid emission factor of 0.757 tCO₂/MWh is to be applied. These conservative factors are used to calculate emission reductions. In order to facilitate adoption of authentic baseline emissions data and in keeping with the principle of "conservativeness," all UCR Indian RE projects shall use the new conservative grid emission factor of 0.757 tCO₂/MWh in their emission reduction calculations for the 2024 vintage year Hence, the same emission factor has been considered to calculate the emission reduction under conservative approach. The parameters applied in the calculation were validated by the verification team. The verification team confirms that all relevant parameters have been sufficiently considered and the values of the parameters are real, measurable and conservative. Parameters monitored ex-post According to the approved methodology ACM0002-Consolidated baseline methodology for grid-connected electricity generation from renewable sources -Version 22.0, the following parameters will be monitored: <table border="1" data-bbox="603 1563 1436 1727"> <thead> <tr> <th>Parameter</th><th>Description</th></tr> </thead> <tbody> <tr> <td>$EG_{PJ,y}$</td><td>Quantity of net electricity generation supplied by the project/plant/unit to the grid in year y</td></tr> </tbody> </table> The values of the parameters monitored were checked against submitted Joint Meter Readings and invoices and were found correct. <u>Meter Details:</u>	Parameter	Description	$EG_{PJ,y}$	Quantity of net electricity generation supplied by the project/plant/unit to the grid in year y
Parameter	Description				
$EG_{PJ,y}$	Quantity of net electricity generation supplied by the project/plant/unit to the grid in year y				

	Meter Type	Meter Sr. No.	Sub-station	Accuracy class	Make	Calibration Details	Calibration Validity
	Main Meter	13194980	Akal	0.5	L & T	19-02-2021	18-02-2026
	Backup Meter	15624836	Akal	0.5	L & T		
There is no calibration delay for the current monitoring period from 01/10/2021 to 31/12/2024. Management system and quality assurance The monitoring plan presented in the PCN complies with the requirements of the applicable methodology. The verification team has verified all parameters in the monitoring plan against the requirements of the methodology and no deviations have been found. The management system and quality assurance procedures have been reviewed by the verification team through document review and interviews with the project participant. The project participant would train all the monitoring staffs are trained against with related requirement; the training guidelines and monitoring manual are saved and verified. The monitoring plan outlines in the PCN includes: - Monitoring Organization - Monitoring apparatus and installation - Calibration - Data collection - Data Management system The submitted calibration certificates were checked and it was confirmed that the calibrations are conducted periodically as specified in the PCN i.e. at least once in 5 years. There was no delay in the calibration during the current monitoring period.							
Findings	CAR 03 was raised and closed successfully. More information presented appendix below.						
Conclusion	The verification team is convinced of compliance of the monitoring plan with the requirements of the monitoring methodology ACM0002-Consolidated baseline methodology for grid-connected electricity generation from renewable sources -Version 22.0. During the remote audit assessment, the verification team interviewed the PP that the						

	monitoring arrangements described in the monitoring plan are feasible within the project design. The monitoring parameter reported in MR adequately represents the parameters relevant to emission reduction calculation. The calibration report ensures the accuracy of the data reported. The number of CoUs generation is calculated based on this accurately reported data. The calculation was done using an excel sheet where all the parameters were reported. The grid emission factor for electricity is considered as per UCR recommendation for Indian project. In the monitoring report, emission reduction calculations are correctly calculated and reported. The monitoring report meets the requirements of UCR project verification requirements.
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Start date, crediting period and duration

Means of Project Verification	The start date and crediting period of project activity was checked based on the commissioning certificate, PCN, MR and other documents provided.
Findings	CAR 01 was raised and closed successfully. More information presented appendix below.
Conclusion	The project has chosen crediting period start date in UCR as 01/10/2021. The crediting period is chosen as 01/10/2021 to 31/12/2024 and the crediting period for the current monitoring period is 01/10/2021 to 31/12/2024.

Positive Environmental impacts

Means of Project Verification	PP has not claimed any separate positive environmental impact. The project being renewable energy project will reduce fossil fuel use through replacement of the same.
Findings	No findings raised
Conclusion	The project is a renewable energy project and reduces the environmental burden by reducing the dependence on fossil fuel-based power plants.

Project Owner- Identification and communication

Means of Project Verification	PCN, communication agreement, MR, commissioning certificate, power purchase agreement.
Findings	No findings raised.
Conclusion	The project owner was identified through a communication agreement signed between project owner and project aggregator. Commissioning certificates and Power Purchase Agreement were also verified and they clearly establish the project ownership. The identification and communication correctly meet the requirement of project verification and UCR project standard. Project owner: Vaayu India Power Corporation Pvt Ltd

Positive Social Impact

Means of Project Verification	Project has provided temporary employment to local people during its
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	installation and commissioning. Also post commissioning some of people have employed permanently and local people were engaged leading to social financial benefit to surrounding. Overall social impact of project implementation is positive on the surrounding area
Findings	CL 02 was raised and closed successfully. More information presented appendix below.
Conclusion	Project has overall positive social impact

Sustainable development aspects (if any)

Means of Project Verification	PP has claimed SDG Goals 7, 8 & 13. SDG 7 is affordable and clean energy and it is verified during remote audit as the project is solar power plant. SDG 8 is decent work & economic growth and is verified by the supporting documents provided. SDG 13 is climate action. These claims were checked on the basis of supporting documents, JMR & invoice, employment of the local people on the project site and emission reduction calculations respectively.
Findings	No findings raised.
Conclusion	The project has the capability to address SDG 7, 8 and 13.

Internal quality control

The verifier confirms that,

- Due professional care has been taken while reviewing the submitted document.
- There is no conflict of interest as the verifier has no other engagement with either the aggregator or project owner directly or indirectly.
- Verification team consists of experienced personnel.

Project Verification opinion

Assessment team conducted documentation review the PCN against the UCR program verification standard version 2.0 and UCR project eligibility criteria version 7.0 and the UCR-PCN-FORM Version 1.0.

It is confirmed that the project activity is a 50.4 MW of large-scale wind power project located at Sipla, Pithla, Kotri, Senag, Barna and Khuri villages of Jaisalmer District of Rajasthan State in India.

The geo co-ordinates of the project activity have been mentioned in sections above. Assessment team performed a remote audit and confirmed that the location described in the PCN is accurate. The verification was performed on the basis of UCR requirements, and host country criteria, as well as criteria given to provide for consistent project operations, monitoring and reporting.

The verification consisted of the following three phases: i) desk review of the PCN, MR and additional background documents; ii) follow-up interviews with project stakeholders; iii) resolution of outstanding issues and the issuance of the final verification report and opinion.

The project correctly applies the approved baseline and monitoring methodology ACM0002-Consolidated baseline methodology for grid-connected electricity generation from renewable sources -Version 22.0.

The monitoring plan provides for the monitoring of the project's emission reductions. The monitoring arrangements described in the monitoring plan are feasible within the project design, and the project participants are able to implement the monitoring plan. Given that the project is implemented and maintained as designed, the project has achieved the emission reductions of 2,20,152 tCO₂eq during the monitoring period i.e. from 01/10/2021 to 31/12/2024.

The review of the project design documentation and the subsequent follow-up interviews have provided assessment team with sufficient evidence to determine the fulfilment of stated criteria. In our opinion, the project meets all applicable UCR requirements. Assessment team thus requests the registration of the proposed UCR project activity.

Appendix 1. Abbreviations

Abbreviations	Full texts
AMS	Approved Methodology for large-Scale CDM project activities
UCR	Universal Carbon Registry
PCN	Project Concept Note
MR	Monitoring Report
t	Tonnes
NGO	Non-Governmental Organization
ISO	International Organization for Standardization
CAR	Corrective Action Request
CL	Clarification Request
GHG	Greenhouse Gas
MWh	Megawatt Hours
CO ₂	Carbon Dioxide
CH ₄	Methane
N ₂ O	Nitrous Oxide

Appendix 2. Competence of team members and technical reviewers

❖ **Mr. Pankaj Kumar** is a seasoned Environment and Climate Change professional with over 19 years of experience in Climate Change Mitigation & Adaptation, Environmental Due Diligence, Disaster Risk Reduction, Climate Finance, and capacity building. As the Managing Director of Enviance Services Pvt. Ltd., Pankaj Kumar leads a validation and verification body for GHG projects and also providing consultancy services in various areas including Climate Adaptation, Mitigation, Sustainability, and more. Previously, Pankaj Kumar served as a Climate Adaptation Expert with Deloitte Touche Tohmatsu India LLP, contributing to the World Bank project on Asset Management, Institutional Effectiveness, and Road Safety in Bihar. Mr. Pankaj also led the Bihar team for the South Asia Climate Proofing and Growth Development (CPGD) – Climate Change Innovation Programme (CCIP), supported by DFID, which aimed to integrate climate change resilience into planning and budgeting across South Asia. With a strong background in environmental projects, Pankaj Kumar has worked with IL&FS Infrastructure Development Corporation and BUIDCO (Bihar Urban Infrastructure Development Corporation) as an Environmental Specialist for WB & ADB funded projects. Additionally, Pankaj has extensive experience in GHG project validation and verification, having led over 300 projects globally while with UNFCCC accredited DoEs and as an external expert for Gold Standard and Global Carbon Council. Mr. Pankaj is an accredited Lead Auditor, Validator, Verifier, and Technical Expert for multiple sectoral scopes by UNFCCC DoE ENVIANCE SERVICES PRIVATE LIMITED Form name/no: F-5.04 Team intimation & Change request form Issue No: 01 Issue Date: 20-01-22 Effective from: 27-09-24 Revision Date: 27-09-24 Revision no: 03 Prepared by: Vidhya M Approved by: Pankaj Kumar Page 4 of 6 and is on the roster of WASH experts of UNICEF. Mr. Pankaj's expertise spans across various standards including CDM, Verified Carbon Standard, Gold Standard, Global Carbon Council, Natural Forest Standard, Riverse and Social Carbon Standard.

❖ **Ms. Ritu Singh** has done Masters in Environmental Science from Central University of South Bihar, Gaya and bachelor of Science in Zoology from Magadh Mahila College, Patna University, India. She has done Masters' research focused on solid waste management during and post covid-19 pandemic and conducted a survey in Medical Colleges of Bihar to study the trends of waste management. She has more than 2 year working experience in True Quality Certifications Pvt. Ltd. (An outsource entity for LGAI Technological Center, S.A. (Spain) "Applus+ Certification") and has been involved in supporting Audit teams for Validation and Verifications of Project Activities (Renewable and non-Renewable projects) under CDM/VCS/GS4GG/GCC programs. Currently, Ritu is engaged as an internal resource with Enviance Services Private Limited, where she is accredited as a Lead Auditor, Validator, Verifier, and Technical Expert for Sectoral Scope/Technical Area 1.2 by Enviance.

❖ **Ms. Swati Mahajan** is graduate in Environmental Engineering from Shivaji University, India and previously worked as an Environment Engineer at Eco Designs India Private Ltd., Pune. She is adept in designing of landfill sites for solid waste management. She also has hands on experience in cost benefit analysis and preparation of DPRs for SWM projects. She also has done a certified course in carbon capture and storage from Edinburg University. Currently working as GHG assessor for projects under various GHG mechanisms like GCC, ICR, UCR and VERRA.

Appendix 3. Document reviewed or referenced

No.	Author	Title	References to the document	Provider
1	NA	Communication agreement		Project Owner
2	NA	Project Concept Note		Aggregator
3	NA	Monitoring report		Aggregator
4	NA	Emission reduction sheet		Aggregator
5	NA	Declaration on avoidance of doublecounting		Aggregator
6	NA	Commissioning Certificates for the solar power plants		Aggregator
7	NA	Power purchase agreement		Aggregator
8	NA	Joint Meter Readings/invoices for the complete monitoring period		Aggregator
9	NA	Calibration certificates for energy meters		Aggregator
10	NA	Equipment purchase order		Aggregator
11	NA	Grid Emission factor recommended for Indian projects by UCR	Upto year 2023 - https://a23e347601d72166dcd6-16da518ed3035d35cf0439f1cdf449c9.ssl.cf2.rackcdn.com/Documents/UCRStandardAug2024updatedVer7_020824191534797526.pdf Year 2024 - https://medium.com/@UniversalCarbonRegistry/ucr-cou-standard-update-2024-vintage-ucr-indian-grid-emission-factor-announced-ddb790cdc603	General project eligibility criteria and guidance UCR standard version 7.0
12	UCR	UCR Program manual version 6.2 UCR COU standard version 7 UCR Verification standard version 2 UCR terms and conditions		Universal Carbon Registry
13	CDM	CDM approved methodology- ACM0002- Consolidated baseline methodology for grid-connected electricity generation from renewable sources -Version 22.0.		UNFCCC

Clarification request, corrective action request and forward action request

Table 1. CLs from this Project Verification

Classification	☐ CAR ☒ CL/CR ☐ FAR	Number:	01
Raised by:	Ms. Ritu Singh	Document Reference	MR
Finding Description		Date:	04/07/2025
- PP shall submit an undertaking for no double counting for current monitoring period and for project activity has neither been registered nor seeking registration under any other GHG programs. - PP shall submit the single line diagram of the project activity. - PP shall submit detailed project report.			
Client/Responsible Party/Project Proponent Response		Date:	05/07/2025
- PP has submitted an undertaking for no double counting for current monitoring period and for project activity has neither been registered nor seeking registration under any other GHG programs. During assessment it was verified that the project has claimed credits for current monitoring period in UCR and there was no double counting and also the project is not registered under any other GHG programs. - PP has submitted the single line diagram of the project activity. - PP wants to clarify that this project was already registered under the Clean Development Mechanism (CDM) with Project ID 5186 and under the Gold Standard (GS) with Project ID 5013. Please refer the details from the same.			
Validation/Verification Team Assessment		Date:	07/07/2025
- PP has submitted an undertaking for no double counting. During assessment it was verified that the project has no double counting for the current monitoring period in UCR. - PP has submitted the SLD of the project activity and the same was verified during assessment. - The project activity was previously registered in CDM and Gold Standard (GS) registries. During assessment, all the details mentioned in UCR MR was cross checked with the details mentioned in PDD and MR of CDM and GS. Hence, CL 01 is closed.			

Classification	☐ CAR ☒ CL/CR ☐ FAR	Number:	02
Raised by:	Ms. Ritu Singh	Document Reference	MR
Finding Description		Date:	04/07/2025
- PP has claimed SDG 8 for the project activity. Supporting documents are to be provided for the same. - PP shall submit the names of the local stakeholders.			
Client/Responsible Party/Project Proponent Response		Date:	05/07/2025
- PP has submitted the supporting documents to justify the claim of SDG 8 for the project activity. - PP has submitted the names of the local stakeholders.			
Validation/Verification Team Assessment		Date:	07/07/2025

1. The employment details of the project activity have been submitted by the project proponent. Upon assessment, it has been verified that the activity contributes toward the achievement of Sustainable Development Goal 8, which promotes sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all. 2. PP has submitted the list of names of local stakeholders. Hence, CL 02 is closed.
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Classification	☐ CAR	☒ CL/CR	☐ FAR	Number:	03
Raised by:	Ms. Ritu Singh			Document Reference	MR
Finding Description				Date:	04/07/2025
PP shall submit the supporting documents of technical specifications of wind turbines.					
Client/Responsible Party/Project Proponent Response				Date:	05/07/2025
PP has clarified that the project is already registered under the Clean Development Mechanism (CDM) with Project ID 5186 and under the Gold Standard (GS) with Project ID 5013. Accordingly, the relevant details have been taken from the registered CDM and GS PDDs. Kindly refer to the same.					
Validation/Verification Team Assessment				Date:	08/07/2025
PP has provided the CDM link for reference of technical specifications of turbines. Since the CDM process involves multiple levels of validation and verification, the verifier accepts the information contained in the Project Design Document (PDD) as accurate and reliable. Hence, CL 03 is closed.					

Classification	☐ CAR ☒ CL/CR ☐ FAR	Number:	04
Raised by:	Ms. Ritu Singh	Document Reference	MR
Finding Description		Date:	04/07/2025
1. PP shall submit the Commissioning certificate of the project activity. 2. PP shall submit the PPA of the project activity.			
Client/Responsible Party/Project Proponent Response		Date:	05/07/2025
1. PP has submitted the Commissioning certificate of the project activity. 2. PP has submitted the PPA of the project activity.			
Validation/Verification Team Assessment		Date:	08/07/2025
1. PP has submitted the commissioning certificate of the project activity. During assessment it was verified that all the dates mentioned in MR are consistent with the submitted commissioning certificates. 2. PP has submitted the PPA of the project activity. During assessment it was verified that the PPA is signed between Vayu India and Rajasthan Government for 20 years. Hence, CL 04 is closed.			

Table 2. CARs from this Project Verification

Classification	☒ CAR ☐ CL/CR ☐ FAR	Number:	01
Raised by:	Ms. Ritu Singh	Document Reference	MR
Finding Description		Date:	04/07/2025
The start date of the crediting period in UCR inconsistent throughout the MR. Correction sought.			
Client/Responsible Party/Project Proponent Response		Date:	05/07/2025
PP has corrected the inconsistencies in the start date of the crediting period throughout the MR as per UCR requirements.			
Validation/Verification Team Assessment		Date:	08/07/2025
PP has made corrections in the start date of the crediting period in UCR throughout the MR. During assessment it was verified in MR version 1.1. Hence, CAR 01 is closed.			

Classification	☒ CAR ☐ CL/CR ☐ FAR	Number:	02
Raised by:	Ms. Ritu Singh	Document Reference	PCN & MR
Finding Description		Date:	04/07/2025
Under section B.8. of PCN and C.10 of MR, Egy,net parameter is inconsistent. Corrective action sought.			
Client/Responsible Party/Project Proponent Response		Date:	05/07/2025
PP has corrected the inconsistency in the Egy,net parameter under Section B.8 of the PCN and Section C.10 of the MR.			
Validation/Verification Team Assessment		Date:	08/07/2025
PP has made correction in Egy,net parameter in section B.8. of PCN and C.10 of MR. During assessment it was verified in PCN and MR version 1.1. Hence, CAR 02 is closed.			

Classification	☒ CAR ☐ CL/CR ☐ FAR	Number:	03
Raised by:	Ms. Ritu Singh	Document Reference	MR
Finding Description		Date:	04/07/2025
In the MR, the term 'estimated' has been used in place of 'actual'. Corrective action sought.			
Client/Responsible Party/Project Proponent Response		Date:	05/07/2025
PP has replaced the term 'estimated' with 'actual' in the MR, as per the requirement.			
Validation/Verification Team Assessment		Date:	08/07/2025
PP has made all the necessary changes in MR. Now the MR reflects word 'actual' instead of 'estimated'. During assessment it was verified in PCN and MR version 1.1. Hence, CAR 03 is closed.			

Project Verification Report

Table 3. FARs from this Project Verification

FAR ID	xx	Section no.	Date: DD/MM/YYYY
Description of FAR			
Project Owner's response			Date: DD/MM/YYYY
Documentation provided by Project Owner			
UCR Project Verifier assessment			Date: DD/MM/YYYY

ⁱ <https://cdm.unfccc.int/Projects/DB/DNV-CUK1315481394.7/view>

ⁱⁱ <https://registry.goldstandard.org/projects/details/878>